Gustavo Roa

• Website: https://gustavo-roa.github.io/

Education

KANSAS STATE UNIVERSITY Manhattan, KS
Ph.D. in Agronomy - Soil Fertility and Nutrient Management 2026
Graduate Certificate in Applied Statistics
Dissertation title: "Advancing Phosphorus Management and Fertilizer Recommendations for Crop Production"
Advisor: Dorivar Ruiz Diaz. Committee: Ganga Hettiarachchi, Nathan Nelson, and Christopher Valh

KANSAS STATE UNIVERSITY Manhattan, KS
M.Sc. in Statistics - Data Science and Analytics track 2026
Report title: "Model Averaging for Estimation of Optimum Rate and Critical Values under Model Uncertainty"
Advisor: Christopher Valh. Committee: Trevor Hefley, Josefina Lacasa, and Dorivar Ruiz Diaz

KANSAS STATE UNIVERSITY Manhattan, KS
M.Sc. in Agronomy - Soil Fertility and Nutrient Management 2023
Thesis title: "Evaluating diagnostic tools for phosphorus and sulfur management in corn"
Advisor: Dorivar Ruiz Diaz. Committee: Ganga Hettiarachchi, Nathan Nelson, and Christopher Valh

UnADM (Open and Distance University of Mexico) Online, Mexico
Master's in Food Security - Agricultural Production 2020
Thesis title: "Application of sulfur in wheat cultivation (*Triticum aestivum*): Its importance for food security"
Advisor: Eber Quintana. Committee: Mariela Gonzalez, Brenda Santiago, and Dafne Orosco

EARTH UNIVERSITY Limón, Costa Rica
Licenciatura in Agricultural Sciences, degree of Agronomic Engineer. 2017
Graduation project title: "Development of a mobile application for the design and sizing of biodigesters"
Advisor: Mildred Linkimer

- Distinguished student, fourth-year & best graduation project presentation

Experience

KANSAS STATE UNIVERSITY Manhattan, KS
Postdoctoral Research Associate September 2026 – Present

- Advise on statistical analysis and the selection of appropriate methods for research projects across diverse areas of agriculture.
- Analyze and interpret complex datasets using applied statistics, statistical modeling, and programming to support research and decision-making.
- Evaluate model performance and synthesize evidence across experiments to develop data-driven recommendations.
- Support the planning and development of research projects, contributing to scientific publications, technical reports, and communication of results to researchers and agricultural professionals.

KANSAS STATE UNIVERSITY Manhattan, KS
Graduate Research Assistant January 2021 – August 2026

- Led multidisciplinary research on phosphorus (P) and sulfur (S) management, including P correlation and calibration, residual P effects, long-term P placement, and P and S interactions.
- Established and managed more than 50 site-years across corn, soybean, wheat, and sorghum in collaboration with farmers and university extension programs; supervised student interns and coordinated field research activities.
- Designed experiments and analyzed multi-site and multi-environment datasets using statistical modeling to evaluate crop responses, treatment effects, and nutrient management strategies.
- Developed reproducible statistical analyses and data visualizations for scientific publications, technical reports, and presentations at professional conferences.

KANSAS STATE UNIVERSITY Manhattan, KS
Research Assistant July 2019 – December 2020

- Assisted in setting up and maintaining fertility trials, testing company products, and overseeing greenhouse studies, including sulfur and nitrogen use with N^{15} isotope tracers.
- Managed sample handling and processing, ensuring accuracy and consistency in data collection.
- Operated and maintained research equipment, including planting, tractor operations, fertilizer, and herbicide application.

SMITHFIELD HOG PRODUCTION Princeton, MO
Production Inter (Ohio International Intern Program) May 2018 – April 2019

- Assisted in daily farm operations, including feeding, monitoring animal health, and maintaining facility cleanliness.
- Advanced in operational certifications and assisted in artificial insemination and farrowing processes, ensuring successful breeding and birthing outcomes.
- Monitored and recorded data on animal growth and health, contributing to improved farm management practices and animal welfare.

Complementary Education

MIT Institute for Data, Systems, and Society (IDSS) through EdX Online
MicroMaster in Statistics and Data Science (in progress, 2027)
Wageningen University & Research through EdX Online
Professional Certificate in Crop Production and Soil Management 2025
Courses 2: Sustainable Food Security: Crop Production and Sustainable Soil Management: Soil for Life
Harvard University through EdX Online
Professional Certificate in Data Science 2024
Courses 9: R Basics, Visualization, Probability, Inference and Modeling,
Productivity Tools, Wrangling, Linear Regression, Machine Learning and Capstone
Wageningen University & Research through EdX Online
Professional Certificate and XSeries in Food Security and Sustainability 2021
Courses 3: Crop Production, Food Access, and The Value of Systems Thinking

Leadership and Activities

- 2025-present: Member of the ASA, CSSA, and SSSA (Tri-Societies) Graduate Student Committee
  - Supports subcommittees on workshops, leadership, webinars, travel grants, and strategic planning for the Tri-Societies.
- 2026: Essential Overlaps for Crop Improvement – Executive Committee Chair
  - Coordinated the overall organization of the symposium and served as moderator during the event.
- 2026: ASA DataFest at Kansas State University – Mentor

- Mentored student teams during the ASA DataFest competition, providing guidance on data analysis, statistical methods, and project development.

- 2023-2025: GSC/SGA Liaison (Graduate Student Council) / (Student Governing Association) – Kansas State University
  - Facilitates communication and collaboration between the Graduate Student Council and the Student Governing Association, advocating for graduate student interests.
- 2024-2025: Graduate Student Representative for College of Agriculture Committee on Planning – Kansas State University
  - Represents graduate students in strategic planning and decision-making within the College of Agriculture.
- 2023-2024: Member of the College of Ag Graduate Student Advisory Committee – Kansas State University
  - Advise on policies and decisions affecting the College of Agriculture, ensuring student needs are considered.
- 2021-2024: Spanish course instructor (Volunteer) – UFM Community Learning Center
  - Teaches Spanish to community members, enhancing cultural exchange and integration.
- 2023: President of Agronomy Graduate Student Association (AGSA) – Kansas State University
  - Led the association in organizing academic initiatives and professional development events for agronomy students.
- 2023: Graduate Student Representative for the Agronomy Course & Curriculum Committee, Department of Agronomy – Kansas State University
  - Ensures graduate student input in curriculum development to meet educational needs in agronomy.
- 2023: Student moderator program at the 78th SWCS International Annual Conference. August 2023, Des Moines, IA
  - Facilitated discussions and moderated sessions at a major international conference on soil and water conservation.
- 2023: Vice-president of the Soil and Water Conservation Society, student chapter – Kansas State University
  - Coordinated chapter activities promoting soil and water conservation practices within the community.
- 2022: Chair of the Scientific Communication Committee, Agronomy Graduate Student Association (AGSA) – Kansas State University
  - Led efforts to enhance scientific communication skills among agronomy students through workshops and seminars.

Service to the Profession

- Judge for the Practices to Reduce Nutrient Loss and Improve Environmental Quality oral competition, Environmental Quality Community, CANVAS, Salt Lake City, UT, 2025.
- Reviewer for the Journal of Soil and Water Conservation, Soil and Water Conservation Society, 2024–2025.
- Student moderator, 78th SWCS International Annual Conference, Des Moines, IA, 2023.

Awards

- 2nd place, Poster competition. Applied Statistics in Agriculture and Natural Resources, Lincoln, NE, 2026
- 3rd place, AgTech Hackathon CANVAS-Bayer Crop Science, Salt Lake City, UT, 2025
- 1st place, NUE Workshop Hackathon Team Competition, Ames, IA, 2025
- North Central Extension-Industry Soil Fertility Conference - Graduate Student Award, Des Moines, IA, 2021
- Distinguished student, fourth year. EARTH University, Guacimo, Costa Rica, 2017
- Best Graduation Project Presentation in Renewable Energies. EARTH University, Guacimo, Costa Rica 2017

Scholarships

- Kansas Corn Next Generation Scholarship Graduate Level Winner (2026). Amount $1,000
- Bonczkowski scholarship and Armbrust scholarship (2024). Amount $1,250
- Kansas Corn Next Generation Scholarship Graduate Level Winner (2024). Amount $1,000
- Agronomy Graduate Student Association Club Scholarship (2021 & 2023)
- Graduate Student Council-Leadership Scholarship (2024). Amount $2,000
- Timothy R. Donoghue Graduate Scholarship (2023). Amount $2,000
- BECAL Scholarship (2021-2023). Amount $90,000
- Irvin D. and Dora Mae Atkins Family (2023). Amount $1,500
- Dr. Neal F. and Florence E. Morehouse (2022). Amount $1,500
- Schrader-Massier Graduate Excellence Fund (2021). Amount $1,500
- Mastercard Foundation Scholarship (2014-2017). Approximate amount $100,000
- Perry Foundation through Fundación Paraguaya (2014-2017). Amount $4,800

Grant proposals

Kansas Fertilizer Research Funds (2021). "Developing updated phosphorus fertilizer recommendations for corn in Kansas". led by Dr. Dorivar Ruiz Diaz. Amount $61,662

Skills & Interests

Technical: Proficient in programming with R, intermediate with SAS and Python. Practical experience using QGIS and GEE.
Language: English – Full professional proficiency; Portuguese – Full professional proficiency; Spanish – Native or bilingual proficiency; Guarani - Native or bilingual proficiency.
Laboratory: Proficient in various phosphorus extraction methods, soil pH and colorimetric, ICP, and NIR analysis.
Interests: Data analysis, applying statistics, food security, soil fertility, nutrient management, plant nutrition and soil science education.

Professional Memberships

- American Society of Agronomy (ASA) since 2019-present
- Crop Science Society of America (CSSA) since 2019-present
- Soil Science Society of America (SSSA) since 2019-present
- Soil and Water Conservation Society (SWCS) since 2022-present
- International Society of Precision Agriculture (ISPA) since 2023-present
- American Statistical Association (ASA) since 2025-present

Publications

Peer-Reviewed Journal Articles

- **Roa, G. A.**, & Ruiz Diaz, D. A. (2026). Assessing corn response to sulfur fertilization in Kansas: Efficiency, optimum rate, and diagnostic tools. Agronomy Journal, 118, e70337. https://doi.org/10.1002/agj2.70337
- **Roa, G. A**., & Ruiz Diaz, D.A. (2025). Evaluation of tissue phosphorus concentrations as diagnostic tool for phosphorus nutrition in corn. *Soil Science Society of America Journal*, 89, 70031. https://doi.org/10.1002/saj2.70031
- **Roa, G.A.**; Quintana-Obregón, E.A.; González-Renteria, M.; Ruiz Diaz, D.A. Increasing Wheat Protein and Yield through Sulfur Fertilization and Its Relationship with Nitrogen. *Nitrogen* 2024, 5, 553-571. https://doi.org/10.3390/nitrogen5030037

Datasets

- **Roa, Gustavo**, 2024, "Dataset for meta-analysis on sulfur fertilization effect on wheat protein and yield", https://doi.org/10.7910/DVN/4RLPP1, Harvard Dataverse, V4

Manuscripts in Progress

- **Roa, G.A.**, Spooner, J., Kerns, A.W., Hefley, T.J. and Ruiz Diaz, D.A. 2026 (Under-review) Bayesian Inference for Agronomic Response Models: A Critical Soil Test Value Application. *Agriculture & Environmental Letters.*
- **Roa, G.A.**, de Oliviera, J. B., Wanithunga, I., Rice, C.W. and Ruiz Diaz, D.A. 2026 (In preparation) Nitrogen-15 Tracer Techniques for Agronomy Research: Practical Guidelines and Case Studies. *Agrosystems, Geosciences & Environment.*

Extension and Outreach Publications

- Sen, S., and **Roa, G.A.** 2026. "Integrating wheat and other small grains into the corn–soybean rotation." CSA News, Vol. 71(9), August.
- **Roa, G.A**, Rutter, E.B., Sohm, C.D., Cominelli, S., Demarco, J., Sivarathri, B.S, and Sumita, S. 2026. "Understanding the foundations of fertilizer recommendations" *CSA News*, Vol. 71(8), July.
- Sumita, S., Sivarathri, B.S, **Roa, G.A.**, and Demarco, J. 2026. "Diversified and environment-based crop rotation" *CSA News*, Vol. 71(6), May.
- Xin, X, **Roa, G.A.**, Nepal, J., Gutierrez, E., Maryam, H. and Ferguson, T. 2026. "Biochar: A promising tool for soil health and nutrient efficiency in sustainable agriculture" *CSA News*, Vol. 71(5), April.
- Seelam, R., Sivarathri, B.S, **Roa, G.A.**, Lambert, A.M., and Joshi, P. 2025. "Scientists' contributions to agriculture: How the world looks without research" *CSA News*, Vol. 71(1), December.
- Xin, X., Nepal, J., de Oliveira, J. **Roa, G.,** Cominelli, S., and Maryam, H. 2025. "Managing soil N for productivity and sustainability: Understanding gains and losses" *CSA News*, Vol. 70(12), November.
- Cruz, P., de Oliveira, J., Barra Netto-Ferreira, J., **Roa, G.A.**, Gadhiparti, V. 2025. "Rooted in Resilience: Regenerative Agriculture and the Future of Food Systems" *CSA News*, Vol. 70(8), August.
- **Roa, G.A.** and Ruiz Diaz, Dorivar A. (2023) "Evaluating the Impact of Long-Term Phosphorus Placement on Corn and Soybean Rotation under Minimum Tillage System" Kansas Agricultural Experiment Station Research Reports: Vol. 9: Iss. 8. https://doi.org/10.4148/2378-5977.8545
- **Roa, G.A.**; Rutter, Edmond B.; and Ruiz Diaz, Dorivar A. (2023) "Evaluation of Plant Tissue Analysis to Assess Phosphorus Nutritional Status for Corn and Soybean" Kansas Agricultural Experiment Station Research Reports: Vol. 9: Iss. 8. https://doi.org/10.4148/2378-5977.8548
- **Roa-Acosta, G.A.** and Ruiz Diaz, D. A. (2022) "Evaluation of Soil Test Phosphorus Extractants and Tissue Analysis for Corn" Kansas Agricultural Experiment Station Research Reports: Vol. 8: Iss. 9. https://doi.org/10.4148/2378-5977.8347
- **Roa-Acosta, G.A.** and Ruiz Diaz, D. A. (2022) "Early-Season Corn Response to Broadcast Pre-Plant Phosphorus Fertilizer Application" Kansas Agricultural Experiment Station Research Reports: Vol. 8: Iss. 9. https://doi.org/10.4148/2378-5977.8348

Thesis

- **Roa Acosta, Gustavo A.** 2026. "Advances in Phosphorus Management and Fertilizer Recommendations for Crop Production" Ph.D. Dissertation, Kansas State University, Manhattan, Kansas.
- **Roa Acosta, Gustavo A.** 2026. "Model averaging for estimation of optimum rate and critical values under model uncertainty". Master of Science, Kansas State University, Manhattan, Kansas.
- **Roa Acosta, Gustavo A.** 2024. "Evaluating diagnostic tools for phosphorus and sulfur management in corn". Master of Science, Kansas State University, Manhattan, Kansas.
- **Roa Acosta, Gustavo A.** 2024. "Application of sulfur in wheat cultivation (Triticum aestivum): its importance for food security" [Spanish]. Master's in Food Security, UnADM, Mexico.
- **Roa Acosta, G.A.** 2017. "Development of a mobile application (App) for the design and sizing of biodigesters" [Spanish]. Graduation Project. EARTH University, Guacimo, Costa Rica.

Conference Posters, Presentations & Proceedings

- **Roa, G.A.,** Vahl, C.I., Hefley, T.J., Lacasa, J. and Ruiz Diaz, D.A. (2026). Model averaging for estimation of optimum rate and critical values under model uncertainty. Applied Statistics in Agriculture and Natural Resources. May 2026, Lincoln, NE.
- **Roa, G.A.,** and Ruiz Diaz, D.A. (2025). Critical Soil Test Phosphorus for Corn: Updates and Method Comparison in Kansas. 55th North Central Soil Fertility Conference. November 2025, Des Moines, IA.
- **Roa, G.A.,** Spooner, J., Hefley, T. and Ruiz Diaz, D.A. (2025). Improving Phosphorus Correlation and Calibration Studies with Bayesian Approaches. CANVAS. November 2025, Salt Lake City, UT.
- **Roa, G.A.,** and Ruiz Diaz, D.A. (2025). Refining Soil Test Phosphorus Recommendations for Corn in Kansas. CANVAS. November 2025, Salt Lake City, UT.
- Leiva, J.C., **Roa, G.A.** and Ruiz Diaz, D.A. (2024). Efficiency of Different Nitrogen Management Strategies on Winter Wheat Production. 2024 Nitrogen Use Efficiency Workshop. August 2024, University of Illinois Urbana Champaign, IL.
- **Roa, G.A.** and Ruiz Diaz, D.A. (2024). Enhancing Phosphorus Nutrient Management in Corn Through Tissue Analysis and Diagnostic Tools. ISPA 16th International Conference on Precision Agriculture. July 2024, Manhattan, KS.
- **Roa, G.A.** and Ruiz Diaz, D.A. (2024). Evaluation of Long-Term Phosphorus Management: An 18-Year Field Study. SSSA Bouyoucos Summer Conference. June 2024, San Juan, Puerto Rico.
- **Roa, G.A.** and Ruiz Diaz, D.A. (2024). Exploring Long-term Phosphorus Management Strategies for Optimizing Crops Yields in Kansas. Great Plains Soil Fertility Conference. March 2024, Lubbock, TX.
- **Roa, G.A.** and Ruiz Diaz, D.A. (2023). Assessing Sulfur Response, Utilization Efficiency, and Diagnostic Tools for Corn in Kansas. 53rd North Central Soil Fertility Conference. November 2023, Des Moines, IA.
- **Roa, G.A.** and Ruiz Diaz, D.A. (2023). Evaluating tissue phosphorus concentration as a diagnostic tool in corn. ASA, CSSA, SSSA International Conference. October 2023, St Lois, MO.
- Bourns, M.A., Nelson, N.O., **Roa, G.A.** and Ruiz Diaz, D.A. (2023). Sustainable Sufficiency: A New Approach to Phosphorus Fertilizer Management. ASA, CSSA, SSSA International Conference. October 2023, St Lois, MO.
- **Roa, G.A.,** Rutter, E.B., and Ruiz Diaz, D.A. (2022). Evaluation of Plant Tissue Analysis to Assess Phosphorus Nutritional Status for Corn and Soybean in Kansas. 52nd North Central Soil Fertility Conference. November 2022, Des Moines, IA.
- **Roa, G.A.** and Ruiz Diaz, D.A. (2022). Corn and Soybean Response to Long-term Phosphorus Placement Under Minimum Tillage System in Kansas. ASA, CSSA, SSSA International Conference. November 2022, Baltimore, MA.
- **Roa, G.A.** and Ruiz Diaz, D.A. (2022). Corn Yield Response and Evaluation of Soil Test Phosphorus Extractants in Kansas. ASA, CSSA, SSSA International Conference. November 2022, Baltimore, MA.
- **Roa, G. A.** and Ruiz Diaz, D. A. (2022). Evaluation of Soil Test Phosphorus Extractants and Tissue Analysis for Corn in Kansas. Great Plains Soil Fertility Conference. March 2022. Virtual.
- **Roa, G.A.** and Ruiz Diaz, D.A. (2021). Corn response to phosphorus fertilization and evaluation of soil test methods in Kansas soils. 51st North Central Soil Fertility Conference. November 2021, Des Moines, IA.
- **Roa, G.A.** and Ruiz Diaz, D.A. (2021). Evaluation of Sulfur and Nitrogen Interaction in Winter Wheat Using Different Fertilizer Sources. ASA, CSSA, SSSA International Conference. November 2021, Salt Lake City, UT.
- **Roa, G.A.** and Ruiz Diaz, D.A. (2021). Early-season corn response to broadcast pre-plant phosphorus fertilizer application in Kansas. ASA, CSSA, SSSA International Conference. November 2021, Salt Lake City, UT.
- **Roa, G.A.**, Samudio, L.F., Bonini, C.A., and Ruiz Diaz, D.A. (2020). Soil microbial properties with changes in surface soil pH in a long-term no-till system. ASA, CSSA, SSSA International Conference. October 2020, Virtual.

Invited Talks and Workshops

- **Roa, G.A.** (2026, September 9). Applied Statistics in R: From Data to Results. Workshop organized and led during the Essential Overlaps for Crop Improvement symposium, Kansas State University, Manhattan, KS
- **Roa, G.A.** (2024, December 5). Comprehensive soil management: research on fertilization, nutritional diagnostics, and prediction [Spanish]. Invited speaker at the *World Soil Day Commemoration*. Organized by IAAS Paraguay; SOPACIS; Fertimax; and the Faculty of Agricultural Sciences – UNA, San Lorenzo Campus, Paraguay.

Translation Contributions

- Murphy, P., Minson, S., Harner, J., George, H., Harvey, M., & Wells, D. (2024). Mantenimiento y protecciónde pozos privados / Private well maintenance and protection [Spanish translation by Gustavo A. Roa & Eduardo Gutierrez Brito]. Kansas State University Research and Extension (MF3666S)
- Murphy, P., Minson, S., Harner, J., George, H., Harvey, M., & Wells, D. (2024). Ubicación segura de pozos de agua privados / Private wells: Safe location [Spanish translation by Gustavo A. Roa & Eduardo Gutierrez Brito]. Kansas State University Research and Extension (MF3667S)
- Murphy, P., Minson, S., Harner, J., George, H., Harvey, M., & Wells, D. (2024). Pruebas de calidad del agua para pozos privados / Water quality tests for private wells [Spanish translation by Gustavo A. Roa & Eduardo Gutierrez Brito]. Kansas State University Research and Extension (MF3679S)
- Murphy, P., Minson, S., Harner, J., George, H., Harvey, M., & Wells, D. (2024). Nitrato y aguas subterráneas / Nitrate and groundwater [Spanish translation by Eduardo Gutierrez Brito & Gustavo A. Roa]. Kansas State University Research and Extension (MF857S)
- Murphy, P., Minson, S., Harner, J., George, H., Harvey, M., & Wells, D. (2024). Cloración de choque de sistemas de pozos de agua privados / Shock chlorination of private water well systems [Spanish translation by Eduardo Gutierrez Brito & Gustavo A. Roa]. Kansas State University Research and Extension (MF911S)
- Murphy, P., Minson, S., Harner, J., George, H., Harvey, M., & Wells, D. (2024). Pruebas de sistemas de agua privados / Testing private water systems [Spanish translation by Eduardo Gutierrez Brito & Gustavo A. Roa]. Kansas State University Research and Extension (MF3655S)
- Defoe, P., Presley, D., & Hettiarachchi, G. (2023). Jardinería en suelos contaminados con plomo / Gardening on Lead-Contaminated Soils [Spanish translation by Eduardo Gutierrez Brito & Review by Gustavo A. Roa]. Kansas State University Research and Extension (MF3166S)
- Martin, S., & Hettiarachchi, G. (2023). Jardinería en terrenos baldíos: Análisis de contaminantes en el suelo / Gardening on Brownfields: Testing Your Soil for Contaminants [Spanish translation by Eduardo Gutierrez Brito & Review by Gustavo A. Roa]. Kansas State University Research and Extension (MF3192S)
- Review of translation from English to Spanish of the K-State Research and Extension publications, “Gardening on Lead-Contaminated Soils” and “Gardening on Brownfields: Testing Your Soil for Contaminants”. March 2023. https://ksre.k-state.edu/tuesday/announcement/?id=88801